

Precedence of Logical Operators

1. $\neg$
2. $\wedge$ and $\vee$
3. $\rightarrow$ and $\leftrightarrow$

In arithmetic, multiplication has a higher priority than addition
 $3 \times 4 + 5$ means $(3 \times 4) + 5 = 17$
Wrong: $3 \times (4 + 5) = 27$.

Precedence of Logical Operators

1. $\neg$ Priority
2. $\wedge$ and $\vee$
3. $\rightarrow$ and $\leftrightarrow$

Example.

- $\neg p \wedge q$ means ($\neg p$) and q
 $\sim(p \text{ and } q)$ is not the same as the above.

Precedence of Logical Operators

1. $\neg$
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Example.

- $\neg p \wedge q$ means

- Better write $p \vee q \wedge r$ as

meaningless, because `and' and `or' have the same priority,
We don't know which one we do first without brackets.

Either write: “(p or q) and r” or “p or (q and r)” = “(q and r) or p”

Precedence of Logical Operators

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Example.

- $\neg p \wedge q$ means
- Better write $p \vee q \wedge r$ as
- $p \rightarrow q \vee r$ means p implies (q or r)
This does not mean (p implies q) or r

1.2. Applications of Propositional Logic

Translating English Sentence.

Express the statements using logical connectives.

Example.

Huiyi can go to the beach tomorrow ^{implies} only if she finishes her homework today and tomorrow it is not raining.

p: Huiyi can go to the beach tomorrow

q: she finishes her homework today

r: tomorrow it is raining.

The statement is “ $p \rightarrow q \text{ and } \sim r$ ”, or you can write “ $p \rightarrow (q \text{ and } \sim r)$ ”

1.2. Applications of Propositional Logic

Translating English Sentence.

Express the statements using logical connectives.

Example.

Huiyi can go to the beach tomorrow only if she finishes her homework today and tomorrow it is not raining.

Example.

The automated reply cannot be sent when the file system is full.
if

Meaning: If the file system is full, then the automated reply cannot be sent.

p: the file system is full, q: the automated reply can be sent.

The logical expression is: $p \rightarrow \sim q$.

1.3 Propositional Equivalence

Definition

A compound proposition that is always true/false, no matter what the truth values of the propositional variables are that occur in it, is called a **tautology**/**contradiction**.

Example. Determine whether each of the following is a tautology or a contradiction.

• $p \vee \neg p$

Case1: When p is T, $(p \text{ or not } p) = (T \text{ or not } T) = (T \text{ or } F) = T$.

Case2: When p is F, $(p \text{ or not } p) = (F \text{ or not } F) = (F \text{ or } T) = T$.

In both cases, $(p \text{ or not } p)$ is T, so it is a tautology.

1.3 Propositional Equivalence

Definition

A compound proposition that is always true/false, no matter what the truth values of the propositional variables are that occur in it, is called a **tautology/contradiction**.

Example. Determine whether each of the following is a tautology or a contradiction.

- $p \vee \neg p$ This is a tautology.
- $p \wedge \neg p$ This is a contradiction.

(p and q), (p or q), ($p \rightarrow q$) , all these can have different truth values, depending on those of p and q. They are not tautology/contradiction.

Definition

Two compound propositions P and Q are called **logical equivalent** if the biconditional statement $P \leftrightarrow Q$ is always a tautology (regardless of P and Q are true or not), in which case we denote

$"P \equiv Q"$, or $"P = Q"$, or sometimes $"P \Leftrightarrow Q"$.

Basically, it means that P and Q always have the **same truth values in all cases**.

Example: $P: p \rightarrow q$, $Q: \sim p \text{ or } q$, we know that P and Q are equivalent.
We can say: $(P \leftrightarrow Q)$ is a tautology.

Example: $P: p \rightarrow q$, $Q: \sim q \rightarrow \sim p$ (note that Q is the contra-positive of P)
We can say: $(P \leftrightarrow Q)$ is a tautology.

Example. (commutative law)

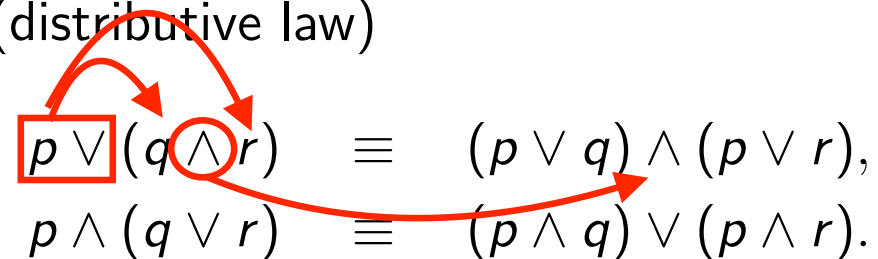
$$p \wedge q \equiv q \wedge p, \quad p \vee q \equiv q \vee p.$$

Example. (associative law)

$$\begin{aligned} p \wedge (q \wedge r) &\equiv (p \wedge q) \wedge r \equiv p \wedge q \wedge r, \\ p \vee (q \vee r) &\equiv (p \vee q) \vee r \equiv p \vee q \vee r. \end{aligned}$$

Note “p or (q and r)” is not equal to “(p or q) and r”

Example. (distributive law)


$$\begin{aligned} p \vee (q \wedge r) &\equiv (p \vee q) \wedge (p \vee r), \\ p \wedge (q \vee r) &\equiv (p \wedge q) \vee (p \wedge r). \end{aligned}$$

Think.

- (Is $\rightarrow$ commutative?) Is it true that

$$p \rightarrow q \equiv q \rightarrow p \quad \text{????} \quad \text{No}$$

Reason: The two statements $(p \rightarrow q)$ and $(q \rightarrow p)$ have different truth values in some cases. (when $p = T$ and $q = F$)

- (Is $\leftrightarrow$ commutative?) Is it true that

$$p \leftrightarrow q \equiv q \leftrightarrow p \quad \text{????} \quad \text{Yes}$$

LHS = $(p \leftrightarrow q) = (p \rightarrow q) \text{ and } (q \rightarrow p)$
= $(q \rightarrow p) \text{ and } (p \rightarrow q) = (q \leftrightarrow p) = \text{RHS.}$

De Morgan's Law

$$\neg(p \vee q) \equiv \neg p \wedge \neg q, \quad \neg(p \wedge q) \equiv \neg p \vee \neg q.$$

Example Use De Morgan's Law to express the negations of the following propositions:

- p : Dina has an iphone and a Samsung smartphone.
- q : Dina has an iphone or a Samsung smartphone.
- r : Irina is sick or she is in school.

$\sim p$: Dina does not have an iphone or does not have a Samsung smartphone.

In practice, to determine whether two compound propositions are equivalent:

- use a truth table; (easy, but not preferred)
- use known logical equivalent statements.

In practice, to determine whether two compound propositions are equivalent:

- use a truth table; (easy and routine, but not preferred)
- use known logical equivalent statements. (preferred method)

Example. (not p unless q) Show that:

$$P : p \rightarrow q \quad \equiv \quad Q : \neg q \rightarrow \neg p.$$

without using the truth table.

$$\text{LHS} = (p \rightarrow q) = (\neg p \text{ or } q) = (\neg p \text{ or } \neg\neg q) = (\neg(\neg q) \text{ or } \neg p) = (\neg q \rightarrow \neg p) = \text{RHS}$$

↑
the equivalence of
p. 19

↑
 $q = \neg\neg q$

↑
commutative law

↑
the equivalence of
p. 19

Definition

A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true.

- When no such assignments exists, that is, when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is unsatisfiable. = **contradiction**.
- When we find a particular assignment of truth values that makes a compound proposition true, this assignment is called a **solution** of this particular satisfiability problem.
- However, to show that a compound proposition is unsatisfiable, we need to show that *every* assignment of truth values to its variables makes it false.

Example. Determine whether each of the following compound propositions is satisfiable? If so, find a solution.

• $P: (p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$

Solve for (p,q,r) (find whether p, q, r is T/F) so that P becomes T.

When $(p,q,r) = (T,T,T)$, then $P = (T \text{ or } F) \text{ and } (T \text{ or } F) \text{ and } (T \text{ or } F) = T \text{ and } T \text{ and } T = T$.
So $P = T$ when $(p,q,r) = (T,T,T)$, then we say P is satisfiable.

Note that P is not a tautology, because $(p,q,r) = (T,T,F)$, then (check it:) P is F.